

Acoustic Analysis of Vowel Quality Using Formant Extraction.

CHOPLIN Alexandre, DUGOUA-JACQUES Maé

Supervisors: M.Lancien

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1 Introduction

1.1 Theoretical Background on Vowel Acoustics

The scientific analysis of speech sounds rests upon the Source-Filter Theory of speech production Nakamura et al., 2011. Within this framework, speech acoustic signals are conceptualized as the output of an acoustic source function passed through a complex frequency-selective acoustic transmission filter. The source function represents the acoustic energy generated at the glottis by the periodic interruption of pulmonary airflow through the self-sustained vibration of the vocal folds. This glottal pulse train determines the fundamental frequency (F_0) of an utterance and establishes a dense harmonic spectrum that drops off in energy at an approximate rate of 12 dB per octave.

The supraglottal vocal tract, spanning from the glottis through the pharyngeal and oral cavities to the labial opening, serves as an acoustic filter. This cavity acts as a time-varying resonator, selectively absorbing or amplifying specific frequencies depending on its physical geometry. The resonant peaks within the vocal tract's acoustic transfer function are designated as formants. Formants represent the natural standing-wave frequencies of the vocal tract configuration for any given articulatory posture. While the fundamental frequency and its harmonics characterize the properties of the glottal source, the spectral envelope, defined by these formants, characterizes the acoustic filter. Vowel quality is determined by the relative frequencies of these formants, primarily the lowest two to three resonances, denoted as F_1 , F_2 , and F_3 .

1.2 Relationship Between Formants and Vowel Quality

The relationship between the spatial positioning of the articulators (articulatory space) and the resulting formant frequencies (acoustic space) is predictable but non-linear. The first formant frequency (F_1) exhibits an inverse relationship with vowel height, which corresponds to tongue body elevation and jaw lowering. When the tongue is high in the oral cavity (as in the high front vowel [i] or the high back vowel [u]), the pharyngeal cavity volume is maximized, creating an acoustic chamber that resonates at a lower frequency. Consequently, high vowels are characterized by low F_1 values (typically between 200 Hz and 400 Hz). Conversely, when the tongue body is depressed and the jaw is lowered (as in the open vowel [a]), the primary constriction shifts toward the pharyngeal region, shrinking its volume and expanding the anterior oral cavity. This articulatory posture results in an elevated F_1 frequency, which can exceed 550 Hz for adult males and 700 Hz for adult females.

The second formant frequency (F_2) is systematically linked to vowel backness, or the anterior-posterior positioning of the major tongue constriction. When the tongue body is advanced toward the alveolar ridge or hard palate, the oral cavity anterior to the constriction is minimized. This shortened front resonator produces high-frequency peaks. Front vowels, such as [i] and [e], are marked by high F_2 values, frequently reaching 2000 Hz in adult males and surpassing 2500 Hz in adult females. When the tongue body is retracted toward the velum or posterior pharyngeal wall, the anterior oral cavity is lengthened, creating a larger resonant chamber that yields a low F_2 frequency. Back vowels, like [o] and [u], show low F_2 values, often dropping below 1200 Hz.

The third formant (F_3) serves multiple functions, prominently signaling the acoustic consequences of lip rounding (labialization). In the French vowel inventory, the presence of front rounded vowels ([y], [ø], [œ]) provides an ideal domain for evaluating F_3 dynamics. Lip rounding structurally extends the physical length of the vocal tract at its anterior opening, which lowers all formant frequencies, but impacts F_3 in front vowels by decoupling internal acoustic cavities. Additionally, F_3 provides critical individual indexical information, reflecting the structural length and anatomical constraints of an individual's vocal tract.

1.3 Structural Hypotheses

1. Biological sex will exert a dominant scaling effect on raw formant values, with female speakers exhibiting higher overall F_1 and F_2 frequencies due to shorter average vocal tract lengths.
2. Age-related differences may appear as subtle shifts in vowel dispersion, centralization patterns, or systemic mergers, reflecting stylistic differences between the generational cohorts.
3. Nasal vowels will exhibit broader acoustic dispersion and higher individual token variance due to the complex interaction of oral and nasal acoustic coupled resonators.

2 Method

2.1 Speakers

The experimental corpus was recorded by four native speakers of French selected to populate a balanced matrix of biological sex and age demographic brackets. The speakers are categorized as follows:

- **S1_Female:** Representative of the younger female demographic, aged in her early 20s.
- **S1_Male:** Representative of the younger male demographic, aged in his early 20s.
- **S2_Female:** Representative of the older female demographic, aged in her early 50s.
- **S2_Male:** Representative of the older male demographic, aged in his early 50s.

None of the participants reported any history of speech, language, or hearing disorders. All speakers exhibited standard regional varieties of Metropolitan French.

2.2 Materials

The stimulus material consisted of ten structurally controlled French sentences designed to contextualize target vowels within balanced phonetic environments. The sentence inventory is detailed below with native LaTeX accent commands to prevent encoding mismatch errors:

1. *Lucile a mis sa belle jupe sur la chaise.* (Targets: [y], [i], [y], [ɛ])
2. *La poule picore la pâte rouge.* (Targets: [u], [i], [a], [u])
3. *Le loup gris passe par la salle vide.* (Targets: [u], [i], [a], [i])
4. *Arthur a vu une puce sur sa tête.* (Targets: [y], [y], [y], [ɛ], [a])
5. *C'est une bête avec de grandes pattes.* (Targets: [ɛ], [a], [ɛ])
6. *La loupe de cette dame est très utile.* (Targets: [u], [a], [y], [i], [ɛ])
7. *Il fume sa pipe toute la journée.* (Targets: [y], [i], [a], [u])
8. *La chute de la tarte sur la table.* (Targets: [y], [a])
9. *Lucas boit une soupe chaude.* (Targets: [y], [a], [u], [o])
10. *La fête se passe dans la rue de la Harpe.* (Targets: [ɛ], [a], [y], [a])

2.3 Recording Conditions

To maintain methodological control, all four speakers were recorded using an identical high-quality microphone interface connected and recorded with a custom Python script via the `sounddevice` library. The technical specifications enforced single-channel (monophonic) recording at a uniform sampling rate (F_s) of 22,050 Hz, satisfying the specific constraints of the project assignment.

2.4 Segmentation Procedure

Acoustic waveforms were segmented and phonetically transcribed using a combined approach involving the automated alignment utility Astali and manual phonetic boundary adjustment via Praat. Vowel offset was established at the point where high-frequency energy in the upper formants (F_1 , F_2) collapsed, or at the initiation of closure for subsequent stop consonants or fricative noise margins. This procedure is subject to criticism due to the subjective nature of the annotator’s guidelines and the judgments of the authors of this report.

2.5 Formant Extraction Settings

Formants were extracted via “fixed_formants_extraction.praat” available at github link. This script is the intellectual property of Mélanie Lancien and was modified for the purposes of this project. This script is fairly general, extracting multiple features that are not all used in this project but may be of interest for similar tasks.

- F0
- F1
- F2
- F3
- F4
- HNR
- Intensity
- Jitter
- Shimmer
- CoG

Feature values are measured every 0.005 seconds and output directly to a .csv file for ease of further processing. This algorithm extracts formants at the acoustic level and requires additional algorithms to perform higher-level analysis (phoneme, word, etc.).

Additionally, the script `formant_quality_verification.py` (also available at github link) was run to verify the accuracy of the extracted formants and remove invalid ones.

3 Results

3.1 Statistical Analysis of Formant Parameters

The automated analysis pipeline processed the segmented tokens to extract descriptive metrics for the primary acoustic formants (F_1 , F_2 , and F_3) along with individual token counts (N). Here, a token is a unit defining a phoneme (specifically a vowel in this context). The complete descriptive output across all speakers is compiled in

Table 1. Vowel labels preserve the string codes used in the backend script data structure, where @ denotes schwa, E denotes the open-mid front vowel [ɛ], E/ denotes front rounded mid-vowels, O denotes the open-mid back vowel [ɔ], and a~ denotes the nasalized low vowel [ã].

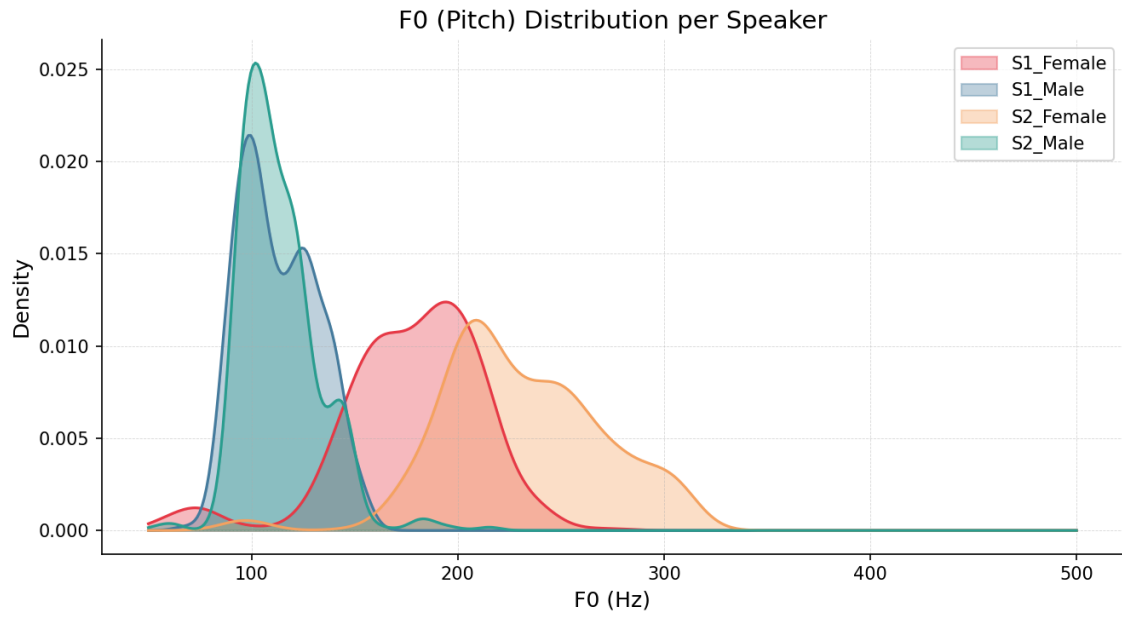


Figure 1: Fundamental frequency (F_0) density distribution curves documenting the non-overlapping profiles of the male and female laryngeal sizes.

Table 1: Comprehensive Acoustic Descriptive Statistics for French Vowels

Speaker	Vowel	N	Mean F_1 (Hz)	SD F_1 (Hz)	Mean F_2 (Hz)	SD F_2 (Hz)	Mean F_3 (Hz)	SD F_3 (Hz)
S1_Female	@	162	410.8	72.5	1497.7	110.5	2522.7	207.1
S1_Female	E	114	417.0	54.6	1714.1	58.9	2577.2	114.0
S1_Female	E/	39	378.7	45.5	1780.2	119.0	2660.8	88.1
S1_Female	O	31	487.7	86.5	1126.8	99.9	2454.3	30.5
S1_Female	a	521	502.7	90.0	1501.9	139.8	2553.4	148.2
S1_Female	a~	43	419.8	95.9	1118.8	183.0	2594.4	275.3
S1_Female	e	7	227.3	42.2	1978.4	112.8	2663.7	33.2
S1_Female	i	153	326.5	58.8	2014.2	152.9	2911.0	233.1
S1_Female	o	20	391.9	40.5	1340.9	48.9	2552.3	156.4
S1_Female	u	93	359.5	49.0	1075.7	191.8	2568.1	180.3
S1_Female	y	205	338.4	53.4	1832.6	212.7	2496.0	157.0
S1_Male	@	150	366.1	71.0	1546.7	138.6	2699.6	243.2
S1_Male	E	133	460.7	51.9	1838.1	71.4	2691.0	114.7
S1_Male	E/	38	399.1	86.8	1900.0	78.4	2708.0	72.9
S1_Male	O	41	534.1	75.3	1187.8	126.0	2721.3	230.0
S1_Male	a	524	579.4	100.2	1538.5	136.3	2689.7	133.8
S1_Male	a~	40	504.2	40.7	989.1	84.1	2825.0	337.4
S1_Male	e	21	370.0	59.0	2073.5	136.1	2660.7	256.0
S1_Male	i	208	293.8	23.3	2007.0	138.2	3160.4	124.2
S1_Male	o	20	394.4	38.3	1301.8	312.7	2998.8	198.4
S1_Male	u	117	332.4	63.5	1051.4	231.4	2859.8	318.0
S1_Male	y	195	330.4	80.2	1838.1	251.2	2672.3	297.3
S2_Female	@	141	378.4	99.1	1588.6	230.1	2507.6	275.2
S2_Female	E	122	464.8	112.7	1826.8	205.5	2592.6	296.1
S2_Female	E/	12	228.7	9.8	1630.0	51.9	2736.4	247.7
S2_Female	O	35	515.5	91.7	1153.4	59.9	2241.0	43.6
S2_Female	a	299	539.8	129.5	1504.1	241.8	2492.4	221.1
S2_Female	a~	19	352.6	93.7	1133.4	103.9	2400.4	187.2
S2_Female	e	5	506.8	37.5	2172.2	25.4	2843.0	143.3
S2_Female	i	125	286.9	61.1	2067.4	248.8	3006.1	371.1
S2_Female	o	11	237.9	7.2	1407.2	72.1	2341.2	90.9
S2_Female	u	89	302.7	65.2	1153.3	200.6	2383.1	141.0
S2_Female	y	262	302.8	64.3	1885.4	228.6	2671.1	289.4
S2_Male	@	161	360.0	76.2	1585.2	214.3	2554.4	218.6
S2_Male	E	77	416.9	44.9	1717.9	147.0	2720.8	359.3
S2_Male	E/	35	304.8	62.8	1883.6	143.3	2676.6	201.5
S2_Male	O	51	464.8	60.3	1208.7	261.6	2548.5	358.2
S2_Male	a	334	493.8	116.9	1419.1	224.0	2729.4	308.6
S2_Male	a~	35	308.0	59.0	1025.6	76.9	2535.8	128.4
S2_Male	e	6	258.3	10.2	1785.7	60.6	2426.7	218.3
S2_Male	i	142	254.3	39.2	2099.1	242.3	3175.4	404.2
S2_Male	o	18	331.1	17.9	1143.2	379.8	2607.7	124.6
S2_Male	u	98	337.0	95.3	1028.0	243.5	2636.5	232.0
S2_Male	y	252	271.9	102.9	1840.1	224.7	2568.2	234.3

To isolate structural comparisons, the spatial boundaries of each speaker’s vowel framework were calculated via a convex hull area integration of category means. The total acoustic area (VSA), expressed in squared hertz (Hz^2), represents the extremes of kinematic articulatory displacements, as compiled in Table 2. “Normalizing and plotting vowels with phonR 1.0.7”, n.d.

Table 2: Calculated Acoustic Vowel Space Area (VSA) via Mean Convex Hull Integration

Speaker	Vowel Space Area (Hz^2)
S1_Female	145,030
S1_Male	205,042
S2_Female	261,864
S2_Male	142,504

To explore internal token distances and the acoustic separation between specific phonemes, Euclidean distances

“Euclidian Distance”, n.d. within the $F_1 \times F_2$ plane were calculated. Table 3 presents an overview highlighting key diagnostic vowel contrasts across the experimental matrix.

Table 3: Pairwise Euclidean Distances (Hz) for Selective Linguistic Contrasts

Vowel Contrast	S1_Female	S1_Male	S2_Female	S2_Male
[i] vs. [u]	939.1	956.4	914.2	1074.3
[i] vs. [a]	541.8	548.7	617.5	720.9
[a] vs. [u]	449.6	546.1	423.4	421.4
[i] vs. [y]	182.0	172.8	182.7	259.6
[a] vs. [ã]	392.0	554.5	415.3	435.2

3.2 Description of Visual and Acoustic Patterns

Analyzing the empirical data tables and corresponding figures reveals clear structural patterns, confirming systematic acoustic differences across all speakers.

3.2.1 Individual Vowel Space Layouts

The individual plots (Figure 2) show that all four speakers maintain the typical asymmetric triangle structure characteristic of the French vowel space. For S1_Female, the high corners are firmly anchored by [i] ($F_1 = 326.5$ Hz, $F_2 = 2014.2$ Hz) and [u] ($F_1 = 359.5$ Hz, $F_2 = 1075.7$ Hz), with the front rounded vowel [y] positioned closely to its unrounded front counterpart, showing an unnormalized Euclidean distance of 182.0 Hz.

For S1_Male, the overall space appears slightly expanded along the lower vertical axis, with the open vowel [a] shifting downward to a mean F_1 of 579.4 Hz. S2_Female demonstrates a highly expanded peripheral array, producing high F_2 values for the front category [i] (2067.4 Hz) along with a lowered boundary for [y] ($F_1 = 302.8$ Hz, $F_2 = 1885.4$ Hz). S2_Male maintains a more compact layout, characterized by lower overall formant positions due to a larger physiological vocal tract.

3.2.2 Distributional Trajectories via Boxplots

The integrated boxplots presented in Figure 3 isolate the frequency variance profiles. For F_1 , the open vowel [a] and the mid-open vowels [ɛ] and [ɔ] display higher median positions across all speakers, whereas [i], [y], and [u] form a low- F_1 cluster. The F_2 distributions clearly separate front vowels ([i], [e], [y], [ɛ]) from back vowels ([u], [o], [ɔ]).

3.2.3 Cross-Speaker Overlays and Normalization

The overlay plots provide a direct view of speaker variation. In Figure 4, the absolute positions of the female systems are shifted upward and to the left relative to the male systems, confirming standard physiological expectations.

The physiological variance disappears when the data are transformed into z-scores via Lobanov normalization (Lobanov normalization explanation(Figure 5)). Once normalized, the structural boundaries of all four speakers align, revealing a shared phonological core with only minor variations in peripheral vowel boundaries.

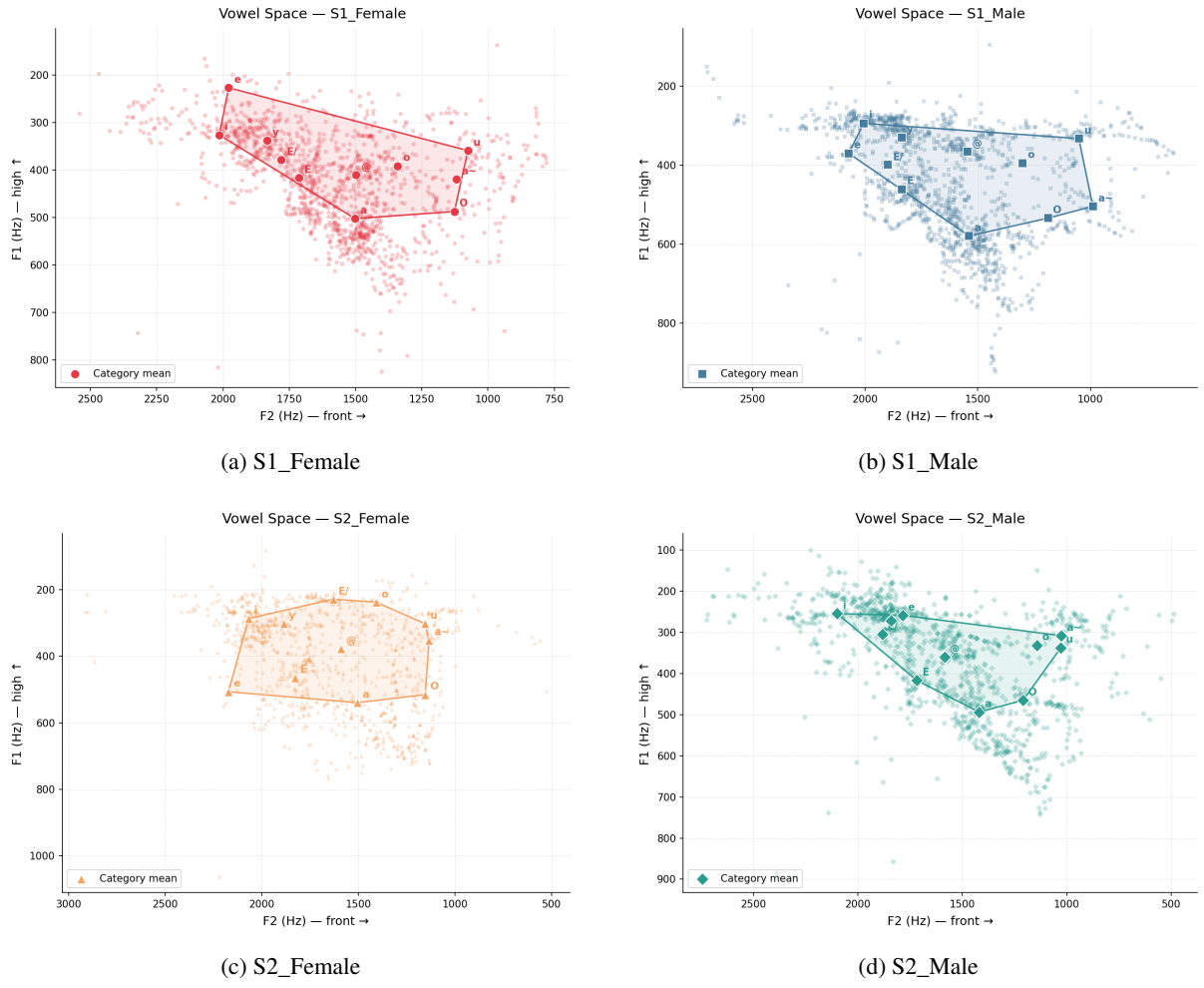


Figure 2: Individual raw $F_1 \times F_2$ acoustic vowel spaces tracking category tokens and spatial boundaries across all four experimental speakers.

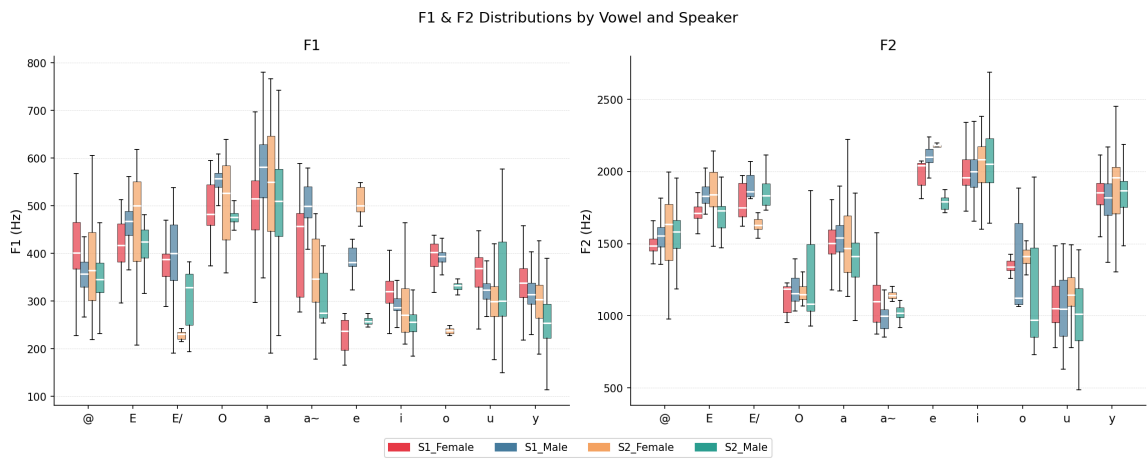


Figure 3: Asymmetric distributions of F_1 and F_2 formant dimensions grouped by target phoneme and colored by individual speaker identity.

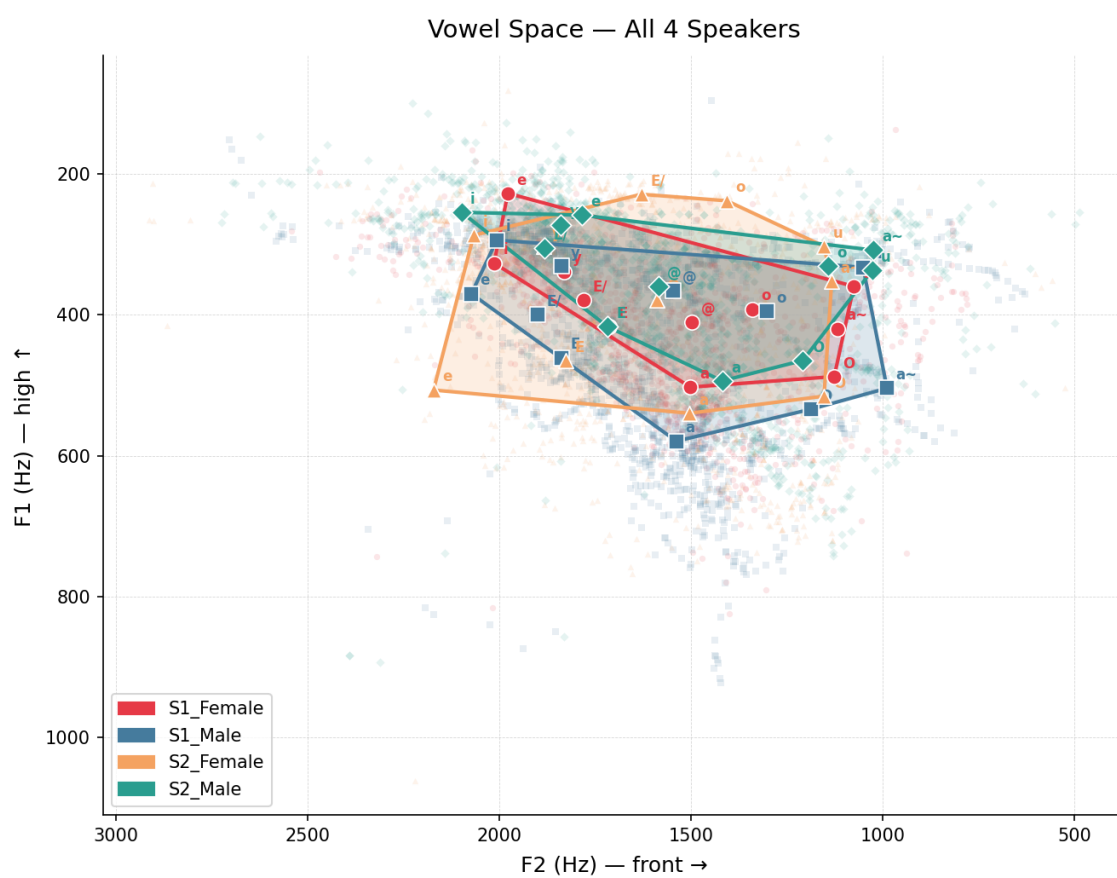


Figure 4: Aggregated unnormalized overlay illustrating absolute cavity scaling variances across all four speakers simultaneously.

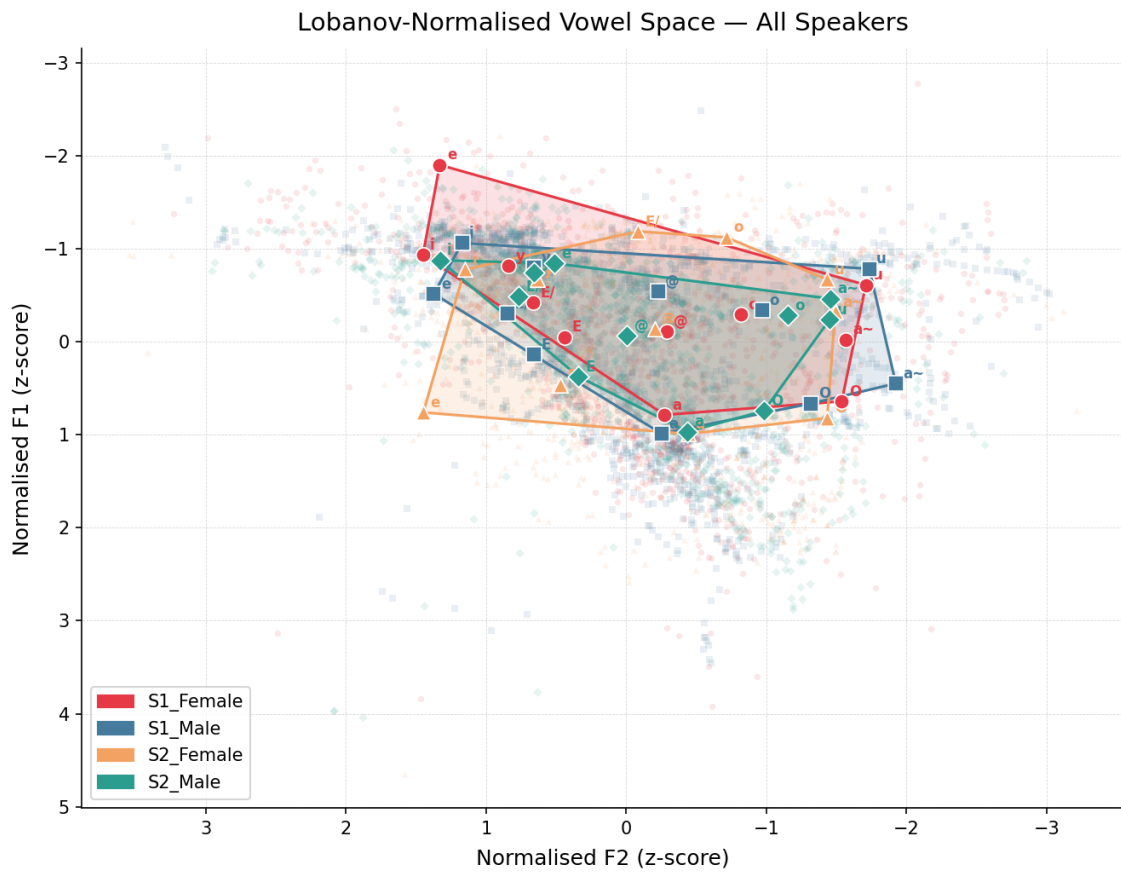
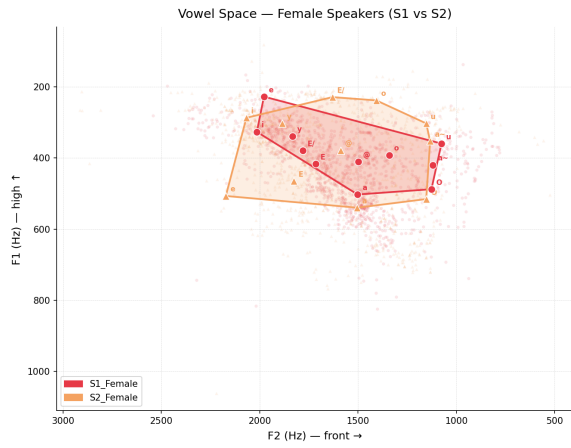
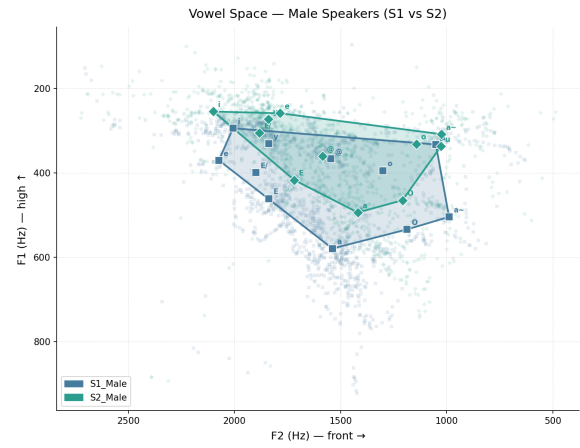


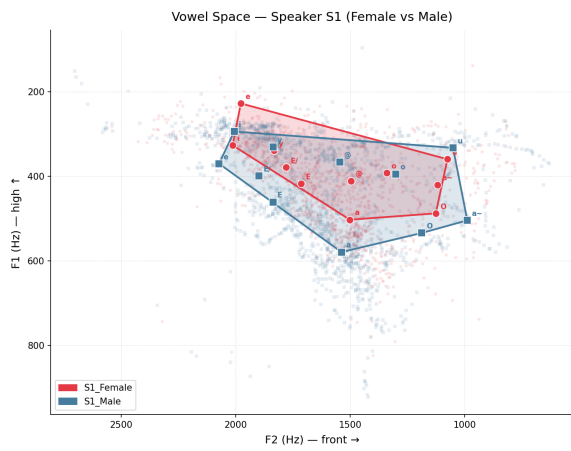
Figure 5: Normalized vowel space overlay combining all participants to demonstrate shared systemic geometry independent of individual anatomy.



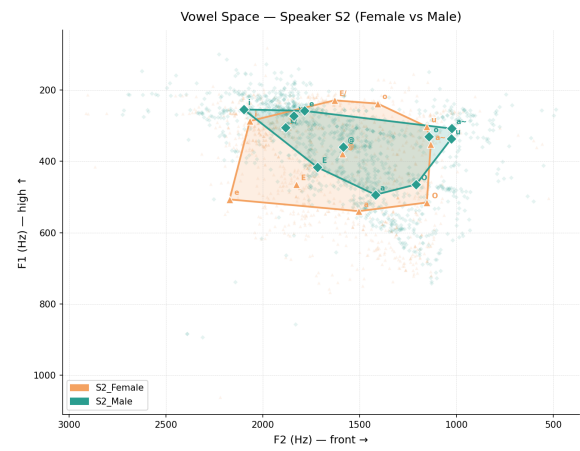
(a) Female Speakers (S1 vs. S2)



(b) Male Speakers (S1 vs. S2)



(c) Speaker S1 (Female vs. Male)



(d) Speaker S2 (Female vs. Male)

Figure 6: Pairwise structural overlays separating generational boundaries (top row) and internal biological sex dependencies (bottom row).

4 Discussion

4.1 Differences Between Speakers: Sex and Age Effects

The empirical data reveal clear differences across the speaker matrix, driven by a combination of physiological constraints. The primary difference is linked to biological sex: the female speakers consistently produce higher raw formant frequencies than their male counterparts. This trend is a direct acoustic consequence of vocal tract scaling. Because human females have shorter average vocal tract lengths than males, their internal resonant cavities are smaller, which systematically shifts their formants higher.

Age-related variation appears as subtle shifts in vowel space positioning rather than changes in the overall inventory. As shown in Table 2, the total acoustic area (VSA) does not follow a simple linear pattern based on age or sex alone.

All differences observed between speakers from different age groups appear to be more closely linked to different speech styles, with some speakers exhibiting more careful articulation than others.

4.2 Effects of Coarticulatory Context

Because the vowels were extracted from naturalistic sentences rather than isolated words, they are highly sensitive to coarticulatory context Ostry et al., 1996. For instance, in Sentence 1 (*Lucile a mis sa belle jupe sur la chaise*), the target [ɛ] in *chaise* is surrounded by the preceding or following voiceless phoneme [z]. The fronted articulation of these coronals tends to shift the F_2 of the vowel higher.

This dynamic articulation explains why the standard deviations for [y] are consistently higher than those for its unrounded counterpart [i]. For example, S2_Female shows an F_2 standard deviation of 228.6 Hz for [y], but only 248.8 Hz for [i]. This increased variance reflects the articulatory adjustments made during connected speech.

4.3 Acoustic Characterization of Nasal Vowels

The nasal vowel [ã] exhibits a distinct acoustic profile. Structurally, the introduction of the nasal cavity into the vocal tract creates nasal antiformants (spectral zeros) that cancel out specific frequencies, while adding extra nasal formants. This complex interaction often lowers the energy of the first formant.

Acoustically, this causes the nasal tokens to scatter over a wider area, as seen in the large standard deviations in Table 1. For S1_Female, the F_2 standard deviation for [ã] reaches 183.0 Hz, which is considerably higher than her variance for oral [a] (139.8 Hz). This expanded distribution reflects the acoustic complexity of nasalization, which makes stable formant tracking more difficult. Lo, 2025

4.4 Measurement Limitations and Formant Tracking Errors

Any acoustic analysis using automated extraction routines must account for tracking errors. This algorithm operates on the assumption that the vocal tract functions as an almost linear system. However, this assumption introduces systematic errors in two main contexts:

1. **High Fundamental Frequency (F_0):** For female speakers with high pitch registers, the harmonic spacing can become wider than the actual bandwidth of the formants. When this happens, the algorithm may misidentify a strong harmonic as a formant peak, or merge F_1 and F_2 into a single broad peak. This issue likely contributed to the high F_2 variance observed in S2_Female’s high vowels.
2. **Nasalization:** The presence of nasal zeros can also mislead the formant extraction algorithm. In vowels like [ã], the algorithm often struggles to distinguish between true oral formants and secondary nasal resonances,

leading to spurious tracking jumps. This limitation is evident in the uneven distribution of nasal tokens visible in the boxplot and facet visualizations.

5 Conclusion

This study provided a comprehensive acoustic analysis of the French vowel space using data from four speakers across two age groups and biological sexes. The empirical results confirm that physiological factors heavily influence raw acoustic metrics, with female speakers displaying higher overall formant frequencies and distinct pitch profiles. The Vowel Space Area (VSA) revealed substantial individual differences in overall vowel space size, with the older female speaker producing the most dispersed vowel space.

Despite multiple attempts to reduce algorithmic and processing approximations, this work faces significant limitations. The analysis also highlighted key limitations of standard speech processing methods, particularly when tracking nasal vowels or speech with high fundamental frequencies. Future work could improve on these findings by incorporating dynamic trajectory tracking across the entire duration of the vowels and utilizing more robust tracking algorithms that are less sensitive to nasal anti-resonances.

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